Binary Search on Answer

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What is Binary Search ?

1. Divide and Conquer
2. Reducing size of search space at every step

$\downarrow$

make the size $= \frac{1}{2}$ of current size

10	20	35	40	50	55	60
0	1	2	3	4	5	6

Give me the index where 55 is present !

① linear search → $O(n)$

② Binary search — if $a[mid] < 55$

— $a[mid] > 25$

$\boxed{25}$

Binary Search Revision

Time Complexity Discussion

a(mid) = val

a(mid) > val

a(mid) < val

We reduce our the size of our search space at every step into half of size of current search space

$$n \xrightarrow{1} n/2 \xrightarrow{1} n/4 \xrightarrow{1} n/8 \cdots 1$$

Recurrence:

- T(n) = T(n / 2) + 1
- T(1) = 1

Time Complexity: O(logn)

$$n \xrightarrow{1} n/2 \xrightarrow{1} n/4 \xrightarrow{1} n/8$$

$$\begin{tikzcd} n \arrow[r] \arrow[r, bend left, "1"] \arrow[rr, bend right, "2"'] \arrow[rrr, bend left, "k"] & n/2 \arrow[r] & n/4 \arrow[r, dotted] & n/2^k \arrow[r, dotted] & 1 \end{tikzcd}$$

$$n/2^k = 1$$

$$2^k = n \longrightarrow k = \log_2 n$$

10 20 5 6 9 11

val = 20

Binary Search Revision

Requirement for using Binary Search

Monotonicity

- f(x) > f(y) iff x > y (increasing monotonic)
- f(x) < f(y) iff x > y (decreasing monotonic)

$x\uparrow$ $y\uparrow$ oR $x\uparrow$ $y\downarrow$

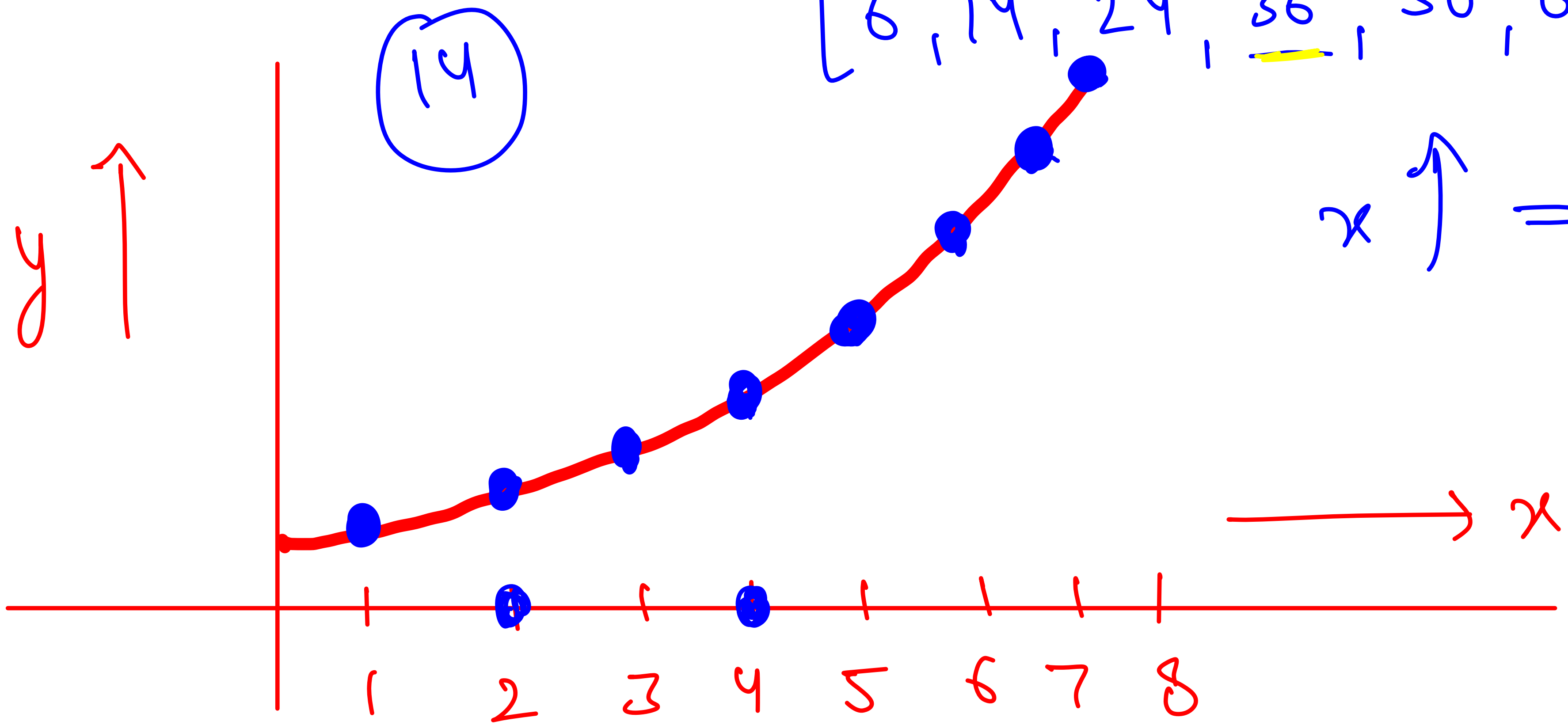
$$f(x) \overset{y}{=} \boxed{x^2 + 5x}$$

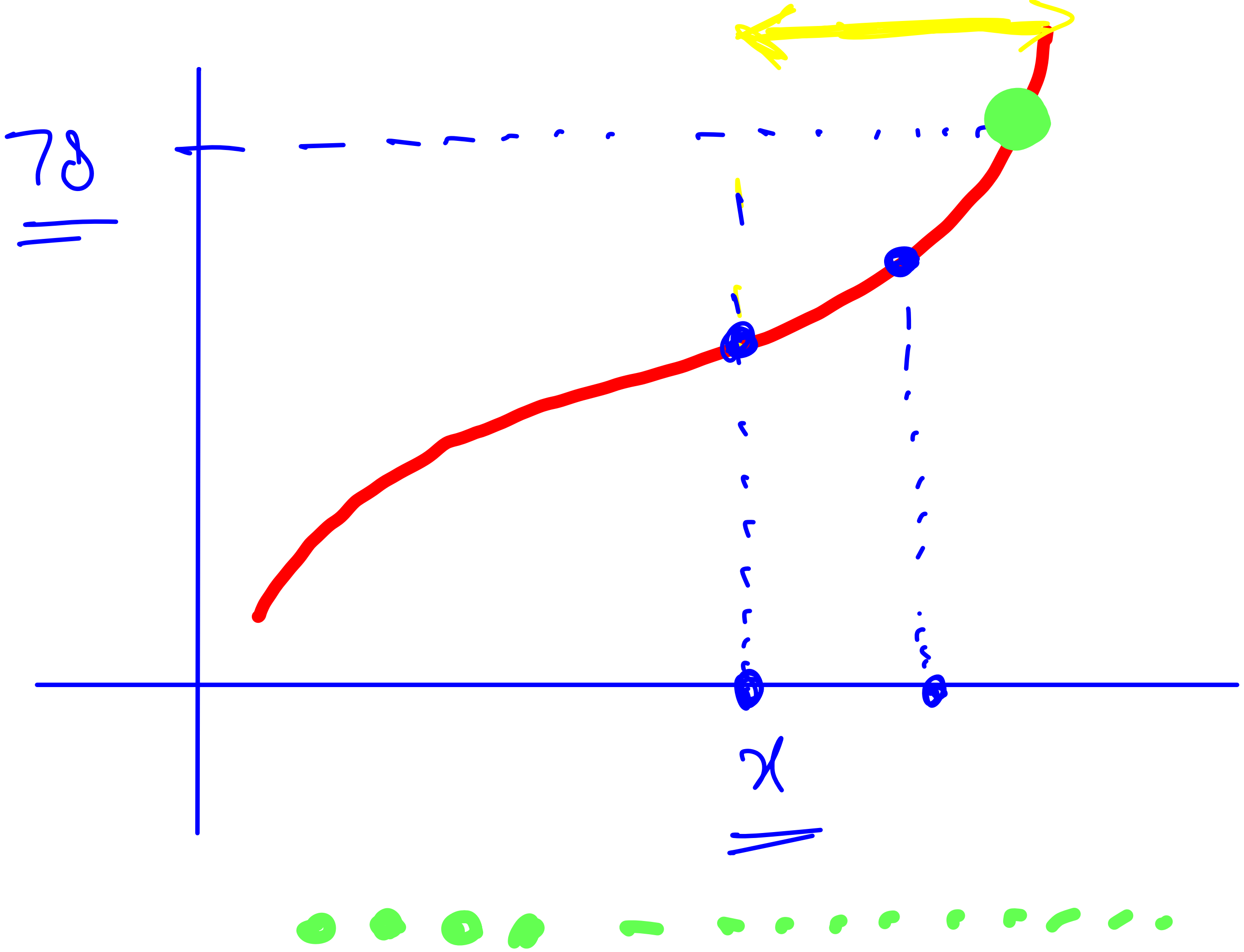
$[[1, 2, 3], 4, 5, 6, 7, 8]$

$\downarrow f(x)$

$[6, 14, 24, 36, 50, 66, 84, 104]$

$x \uparrow \Rightarrow y \uparrow$

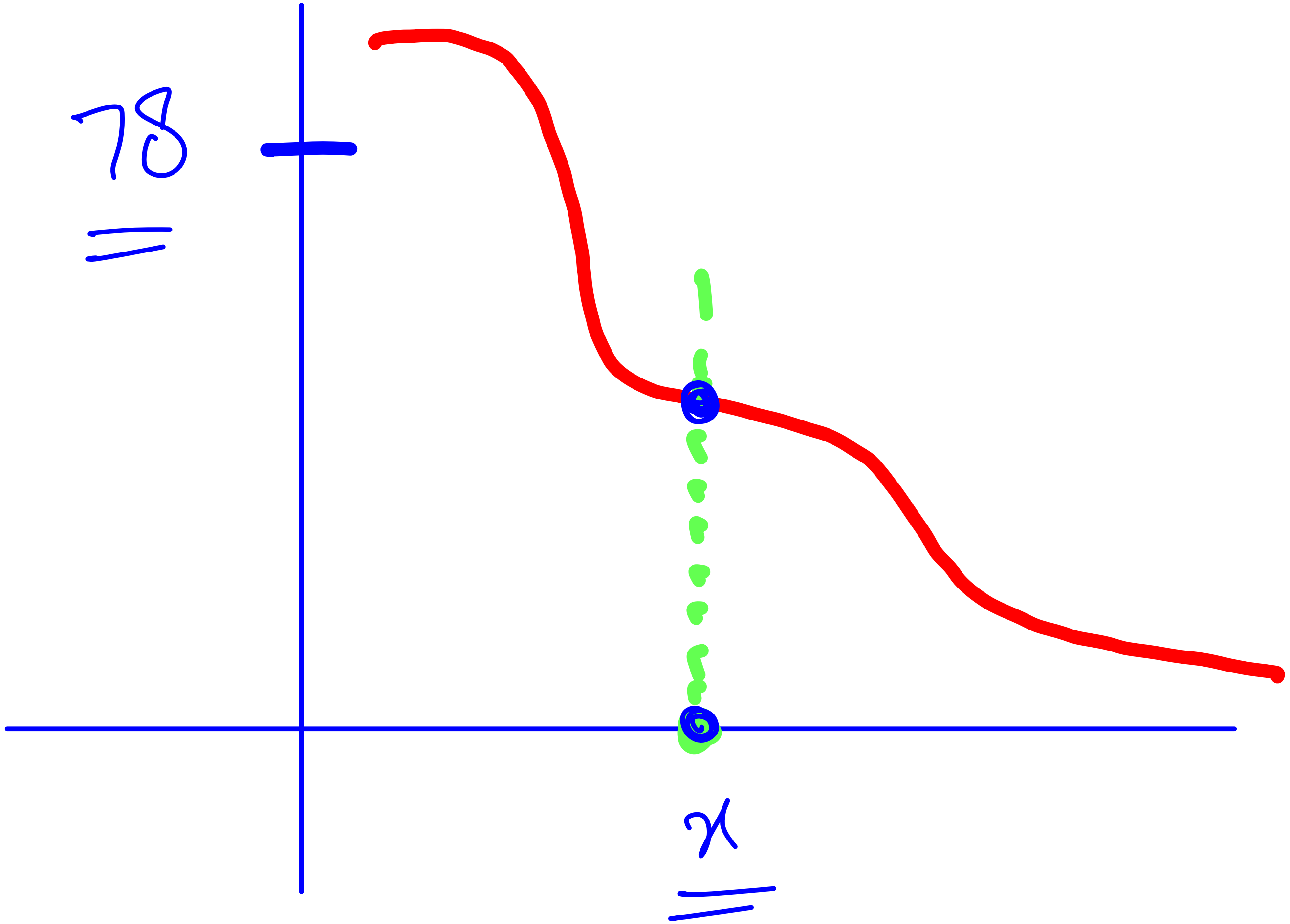




find out for what input $f(x) = 78$

Can we do B.S.

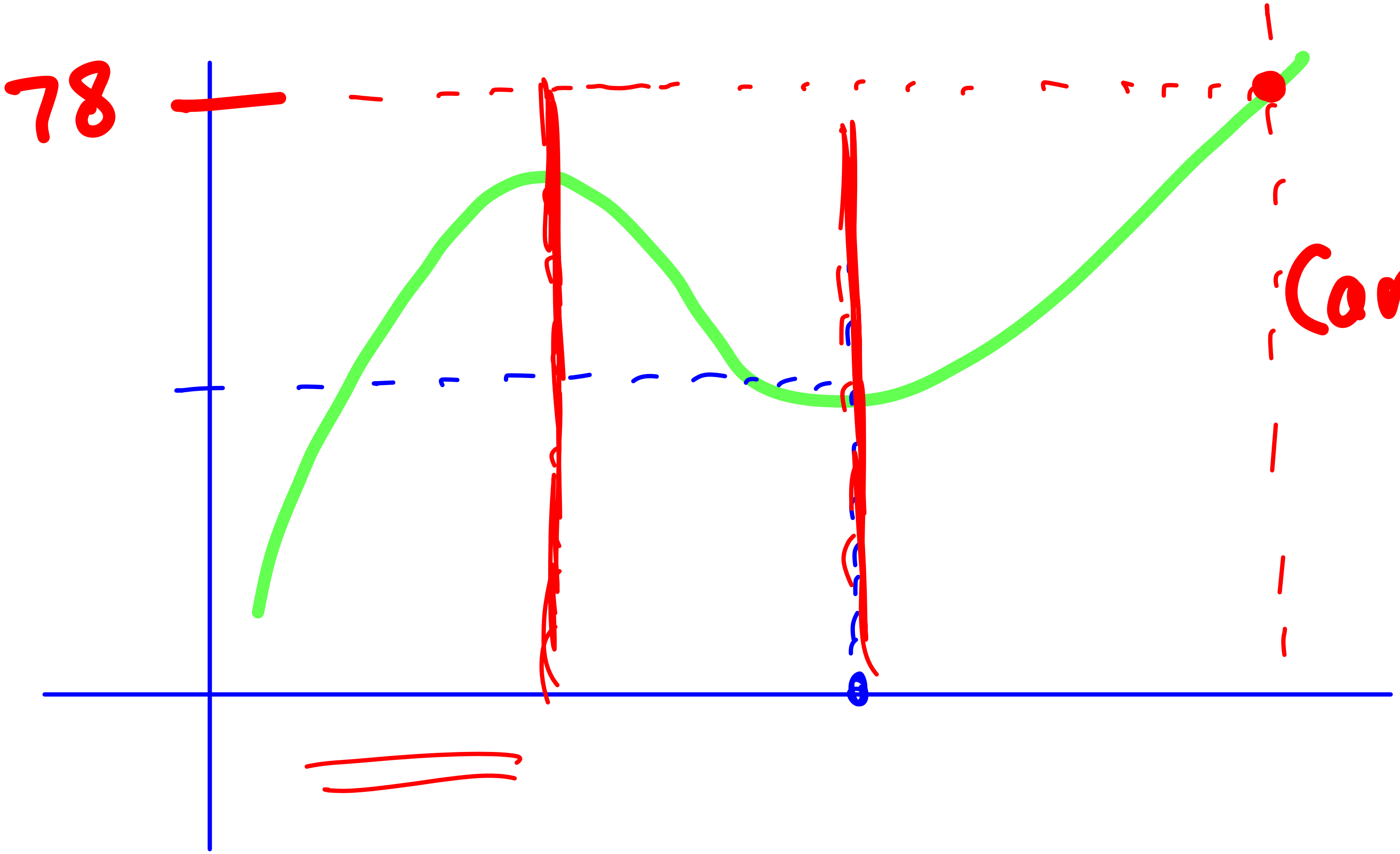
Yes



$x \uparrow \Rightarrow y \downarrow$

Can we do B.S.

Yes



Can we do B.S

No

$10 \quad 11 \quad 17 \quad 18 \quad 20 \quad 22 \quad 30 \quad 50 \quad 60 \leftarrow$ output

$0 \quad 1 \quad 2 \quad 3 \quad 4 \quad 5 \quad 6 \quad 7 \quad 8 \leftarrow$ input

$$\underline{\underline{f(x)}} = a[x]$$

Binary Search Revision

Let's try out a problem!

Given a sorted array on N elements, find out the indices such that

L <= a[i] <= R

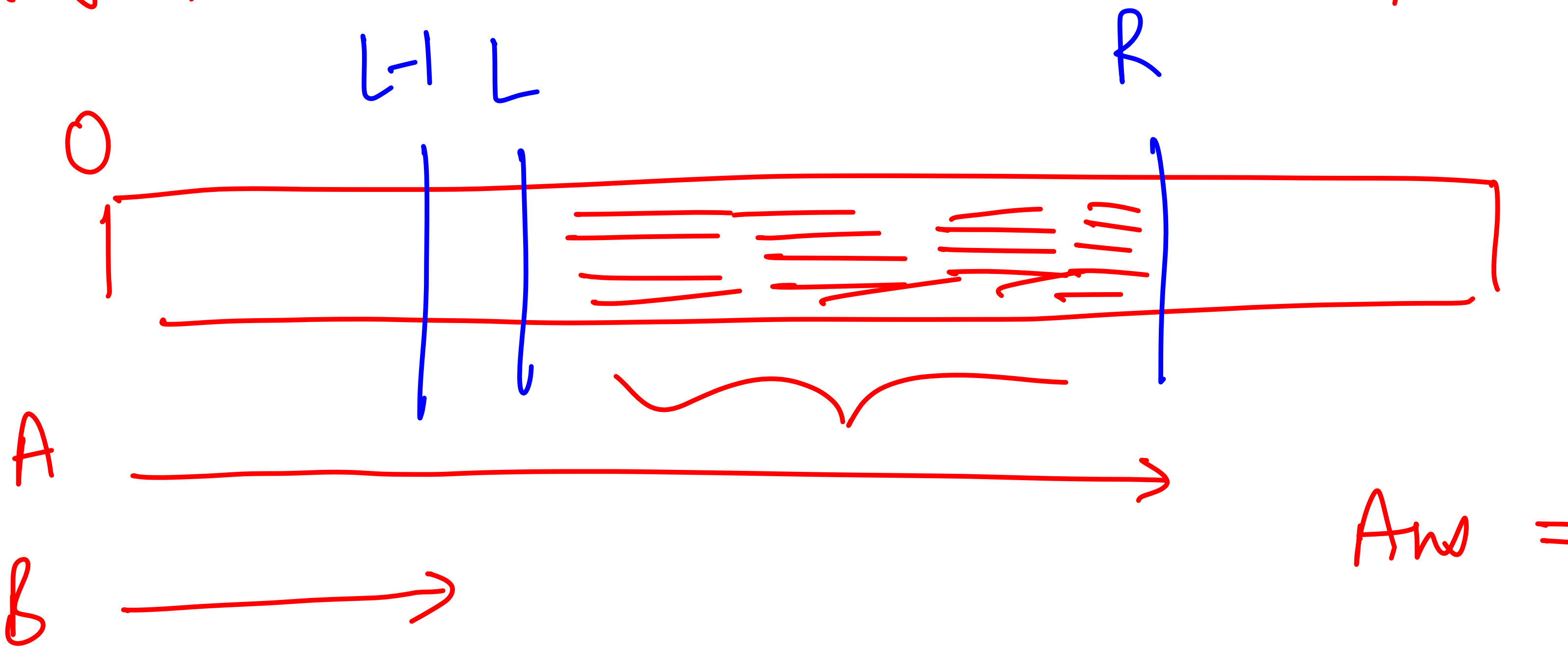
Example: if L = 4 and R = 12, number of elements in the array below satisfying the constraint are 6

1	2	3	5	6	7	8	10	11	16	20

No. of indices (i) such that :

$$L \le a[i] \le R$$

$a[i] \ge L$ ① $a[i] \le R$



Ans = A - B

10	20	22	30	35	40	41	46	48	60

$L = 30$ $\quad R = 45$

1

A $\longrightarrow$ $a[i] \le R$ $\quad O(\log n)$

2

B $\longrightarrow$ $a[i] \le L-1$ $\longrightarrow$ $O(\log n)$

$\boxed{a[i] \le x}$

$Ans = A - B$

$a[i] \leq x$

0	1	2	3	4	5	6	7
10	20	30	40	50	60	70	80

$x = 42$

$ans = 4$

$O(\log_2(n))$

$x = 65$

$ans = 4$

$ans = \max(ans, 5+1) = 6$

Predicate Functions

Predicate functions are functions that return a single **True** or **False** for every input. We use predicate functions to check if our input meets some condition or not.

$f(x) = T$ if x is prime

Examples:

- F(x) = True if x > 10 otherwise False
- F(x) = True if x is a character otherwise False
- F(x) = True if x^2 is an odd number otherwise False

T F

$x \longrightarrow 1 \implies x^2 = 1$

$x \longrightarrow 2 \implies x^2 = 4$

$x \longrightarrow 3 \implies x^2 = 9$

patterns

when x is odd

x^2 is odd

when x is even

x^2 is even

1. odd $\times$ odd = odd

odd $\times$ even = even

even $\times$ even = even

When x is odd then x is of the

form $\boxed{2k+1}$

$x = 5$

$x = 2 \cdot 2 + 1$

$(2k+1)^2 = 4k^2 + 4k + 1$

even even 1

$x = 19$

$x = 2 \cdot 9 + 1$

if x is even $\Rightarrow$ $2k$ $(2k)^2 = 4k^2 \longrightarrow$ even

Monotonic Predicate Functions

$T \rightarrow 1$

$f \rightarrow 0$

TTTTTTTTTFFFFFFF

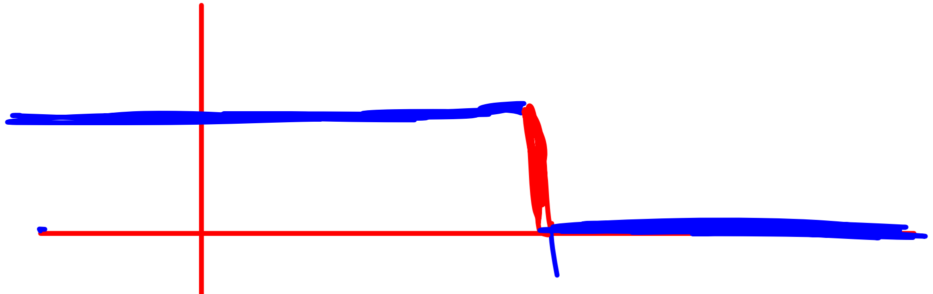
T	T	T	T	F	F	F	F	F

FFFFFFFFTTTTTTTTT

F	F	F	F	F	F	T	T	T

Examples of monotonic predicate functions

① $f(x) \begin{cases} T & \text{if } x < 10 \\ f & \text{o/w} \end{cases}$



② $f(x) \begin{cases} T & \text{if } x^2 + 2x + 7 \leq 10002 \\ f & \end{cases}$

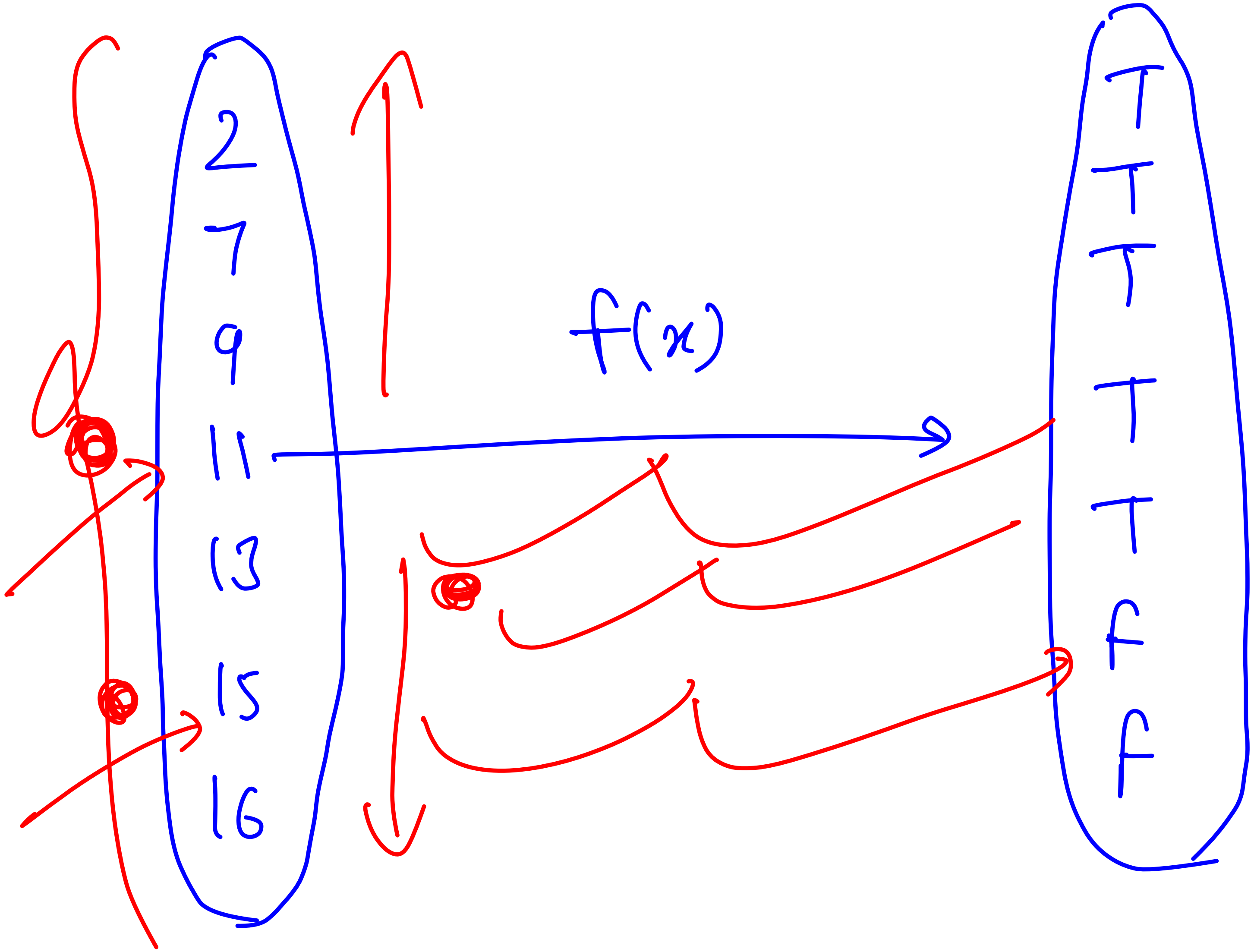
provide $x \geq 0$

Binary Search on Answer

Consider a **monotonic predicate function** P (TTTTFFFFFF) defined over an **ordered set** S (the search space). The search space consists of candidate answers to the problem. Let's suppose we want to find out the highest candidate in the ordered set which returns true when passed to the predict function. Can this be done quickly without trying all candidates?

Example:

- **Ordered set**: {1, 2, 4, 9, 13, 15, 16}
- **Predicate function**: F(x) = True if x - 2 ~~>=~~ <= 10 and false otherwise
- Can you find out the biggest number which returns True


2
7
9
11
13
15
16
f(x)
T
T
T
T
T
F
F

Problem 1: Fair Workload

Given an array of workloads, split it among ‘k’ workers, such that the maximum work that any worker has to do is minimised (can’t change order of workloads).

Eg. [10, 20, 30, 40, 50, 60, 70, 80, 90]

- Solution : 10 20 30 40 50 | 60 70 | 80 90

  First worker - 150,

  Second worker - 130,

  Third worker - 170

minimised maximum workload

Is it possible to partition workload in a way that the highest workload of any worker is less than 170?

10	20	10	10	20	10

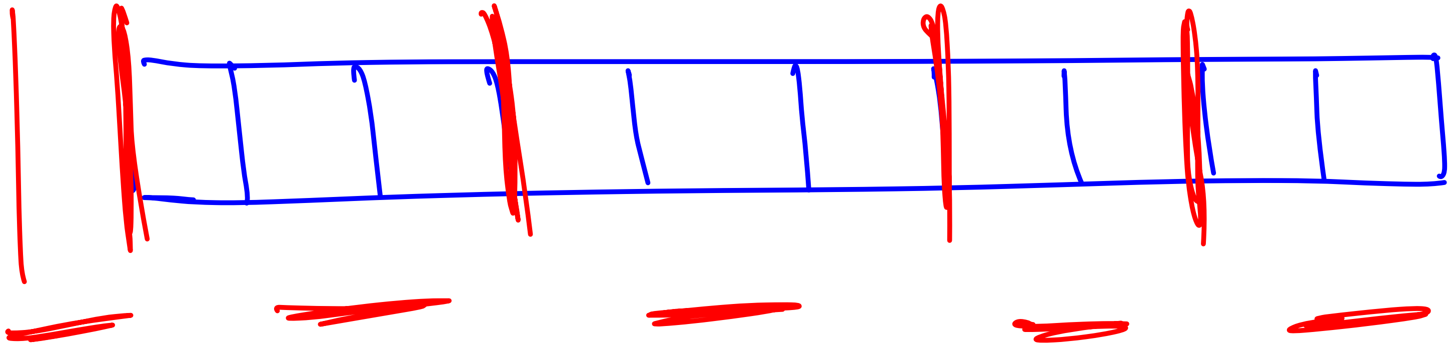
0,80 | 10,70 | 30,50 | 40,40 | 50,30 | 70,10 | 80,0

is there a way to make a split such that

1. both worker do $\leq$ 60 work $\longrightarrow$ T
2. both worker do $\leq$ 30 work $\longrightarrow$ F

k workers

$= 5$



there is a way to split such that all workers do ≤ 100 work

there is no way to split such that all of them do ≤ 20 work min = max element in the array

max = sum of all workloads

10	5	15	100	500	600	7	8

k = 100

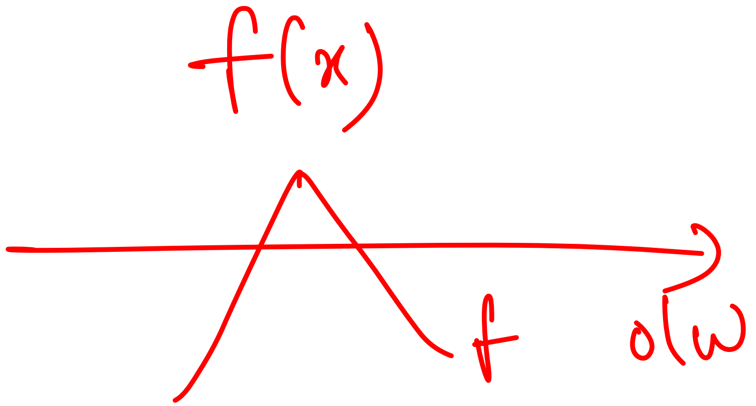
Ans = minimised maximum workload that any worker has to do

600

200

⋮

1000



f
f
f
f
f
f
f
T
T
T

T if there is a way to split the work into k workers such that every worker does $\leq x$ work

$f(x)$ → T if it is possible to split the work into k workers such that all of them do $\leq x$ work

→ F o/w

x

10	20	30	10	10	20	30	20
1	1	1	1	2	2	2	3

$k = 3$

$x = 30 \rightarrow f$

$x = 50 \rightarrow f$

$x = 70 \rightarrow T$

Problem 2: Sqrt(x)

Next class